

lab6

2026-02-25

Q1

a + b

```
## [1] 0.4444445 0.4444444 0.1111111 3.7000000 0.6000000 2.5000000 0.2549510
## [8] 0.3605551 0.0000000

## [1] 0.4444445 0.4444444 0.1111111 3699.9999753 600.0000017
## [6] 2500.0000000 254.9510311 360.5551318 0.0000000
```

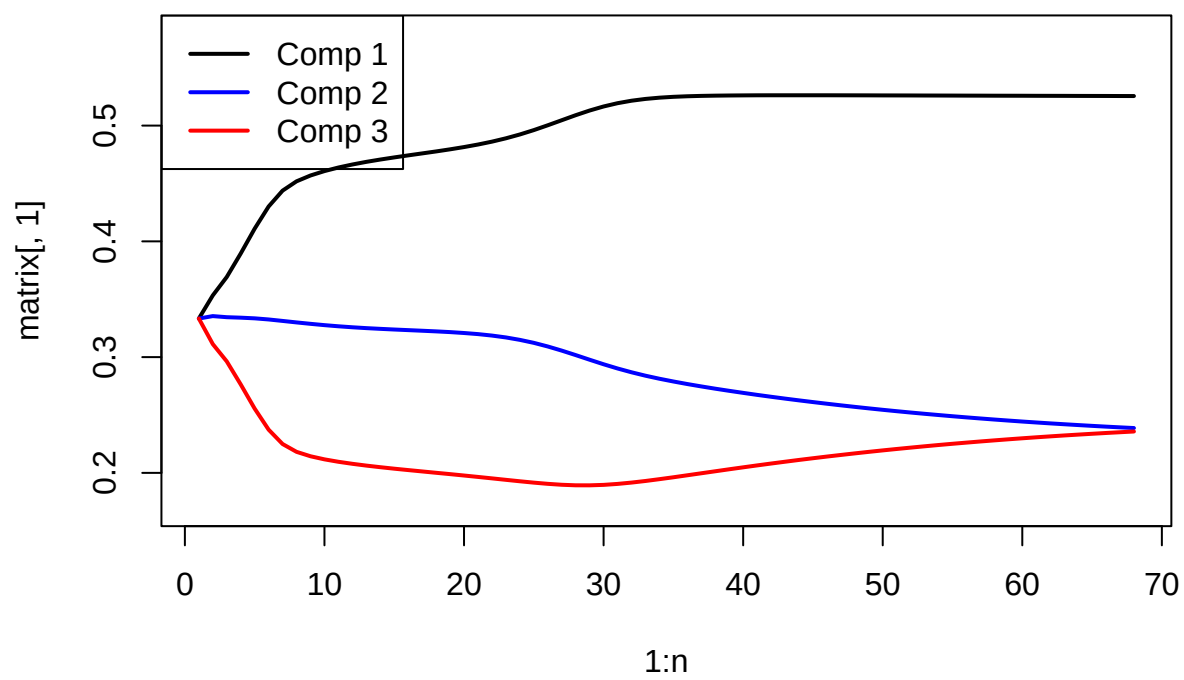
The parameters are on such different scales, using cosine similarity if mul changes by 10 it is going to overpower a change of 0.01 in p2 . $\cos(v_1, v_2)$ denotes the cosine of the angle between v_1 and v_2 . $1 - \cos$ as $\cos = 1$ when the vectors are perfectly aligned.

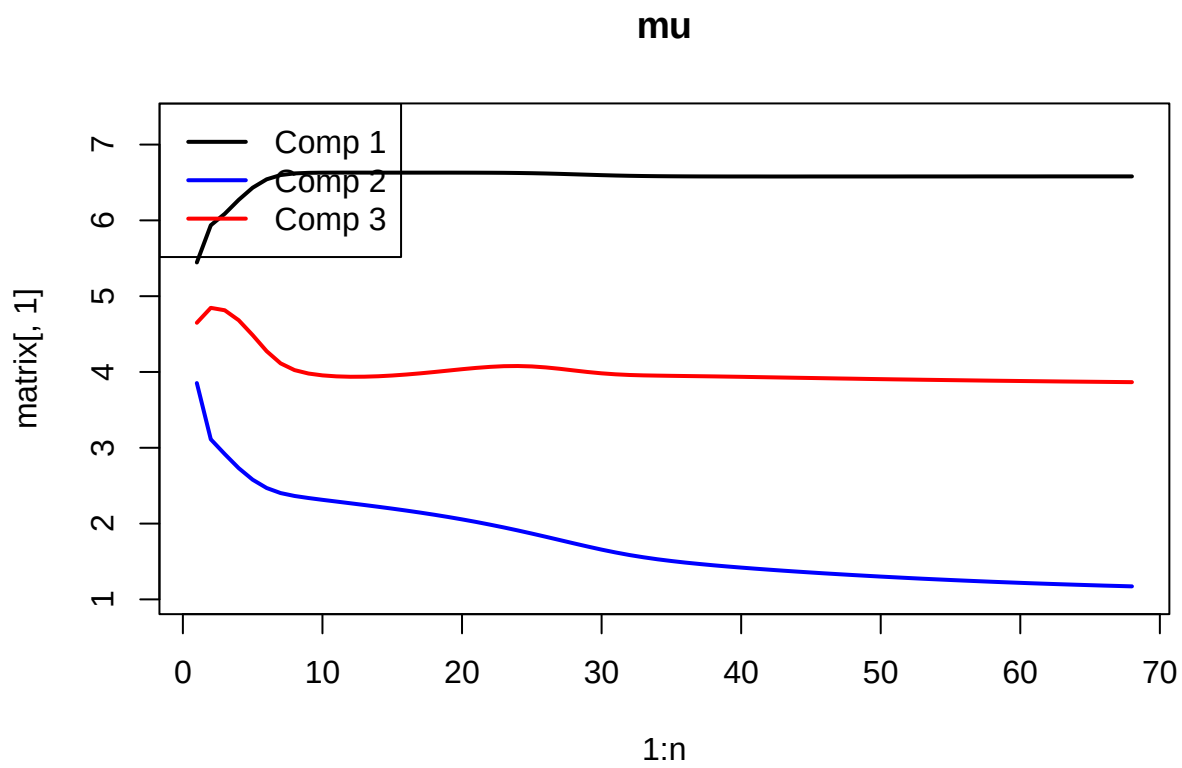
$$cc = \frac{(1 - \cos(v_p, v_{1_p}), (1 - \cos(v_\mu, v_{1_\mu}), (1 - \cos(v_\sigma, v_{1_\sigma}))}{3}$$

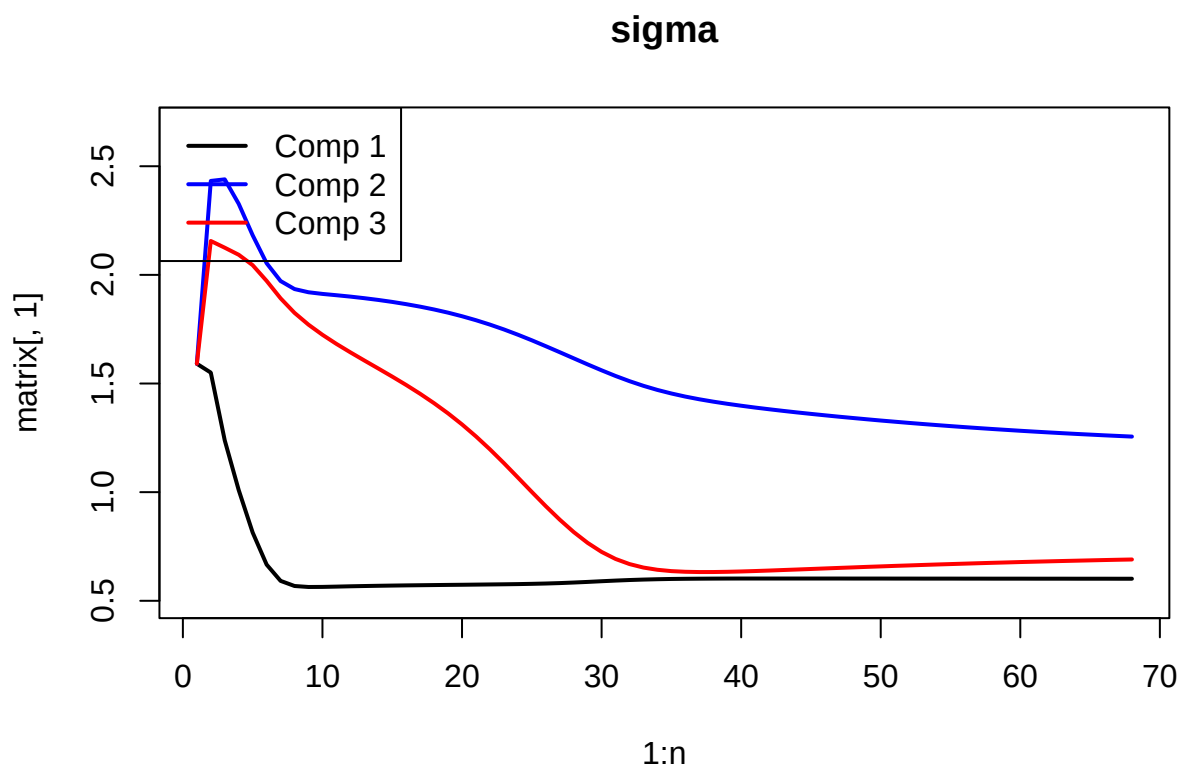
```
## [1] 0.5255415 0.2387176 0.2357409 6.5803106 1.1718721 3.8658191 0.6015409
## [8] 1.2558208 0.6901542
```

It looks like they converged

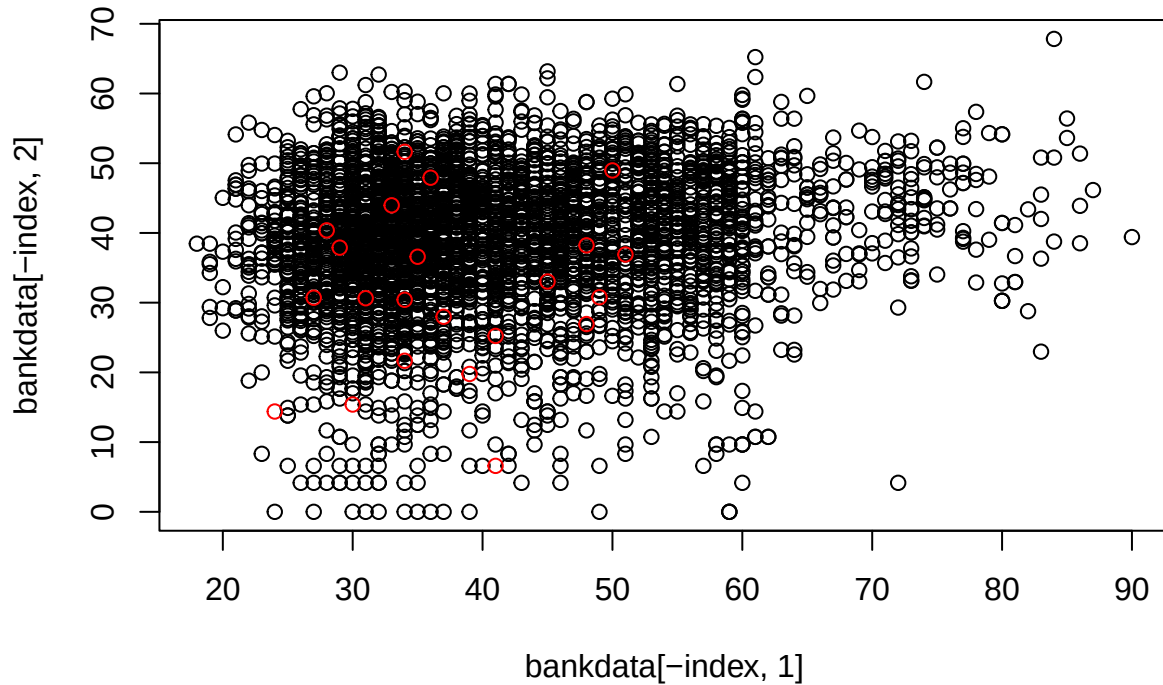
pi







Q2



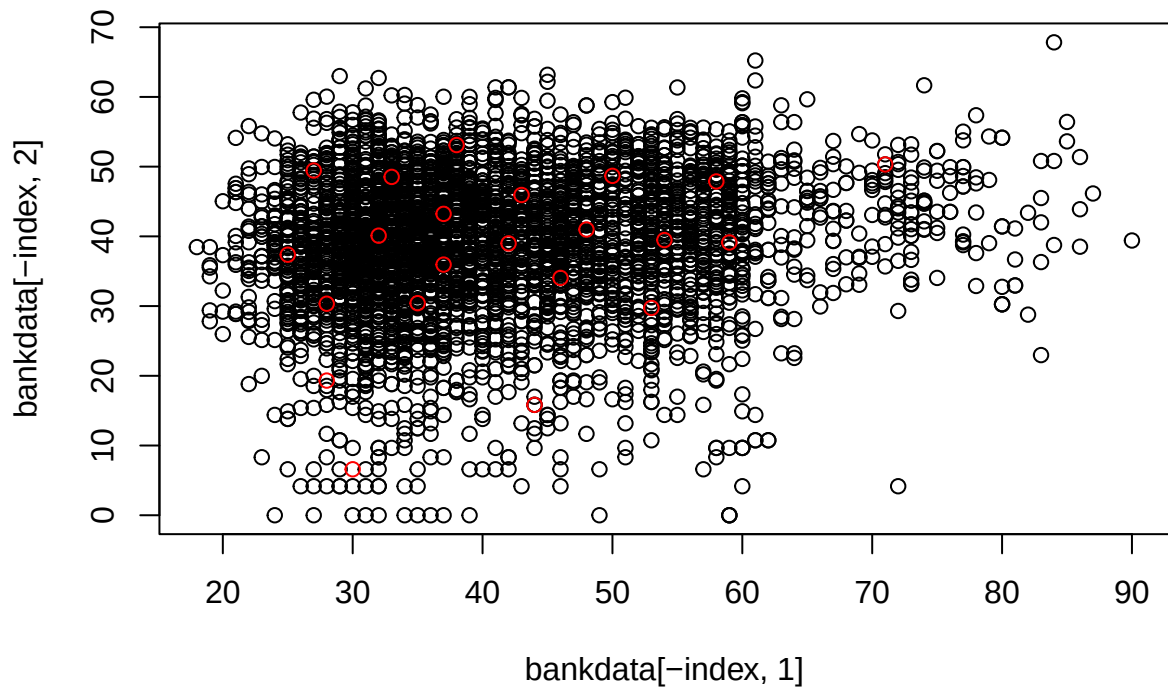
The proposal is a uniform mask, for each individual we swap them out with probability `1-prob_keep`.

`SA(prob_keep = 0.5, alpha = 0.9, beta = 1.02, mj = 50, temp = 10000, it = 200)` This gives 16203 as minimum function value but is painfully slow so cannot run it in the knit.

Lower function value and faster, higher starting temperature and lower alpha makes it decay faster, 0.9 probability swaps 2 people on average.

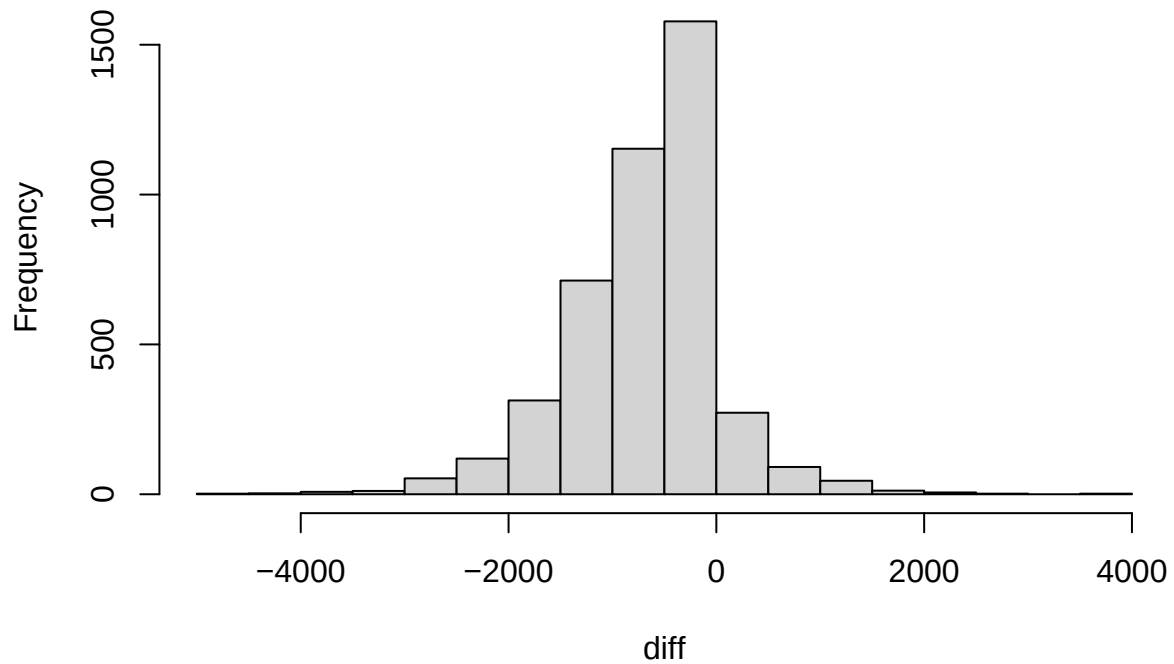
```
## [1] 10
## [1] 20
## [1] 30
## [1] 40
## [1] 50
## [1] "Starting function value:"
## [1] 21936.02
```

Best individuals tested



```
## [1] "Best function value:"  
## [1] 15982.41  
## [1] "Average diff:"  
## [1] -651.12
```

Delta in function

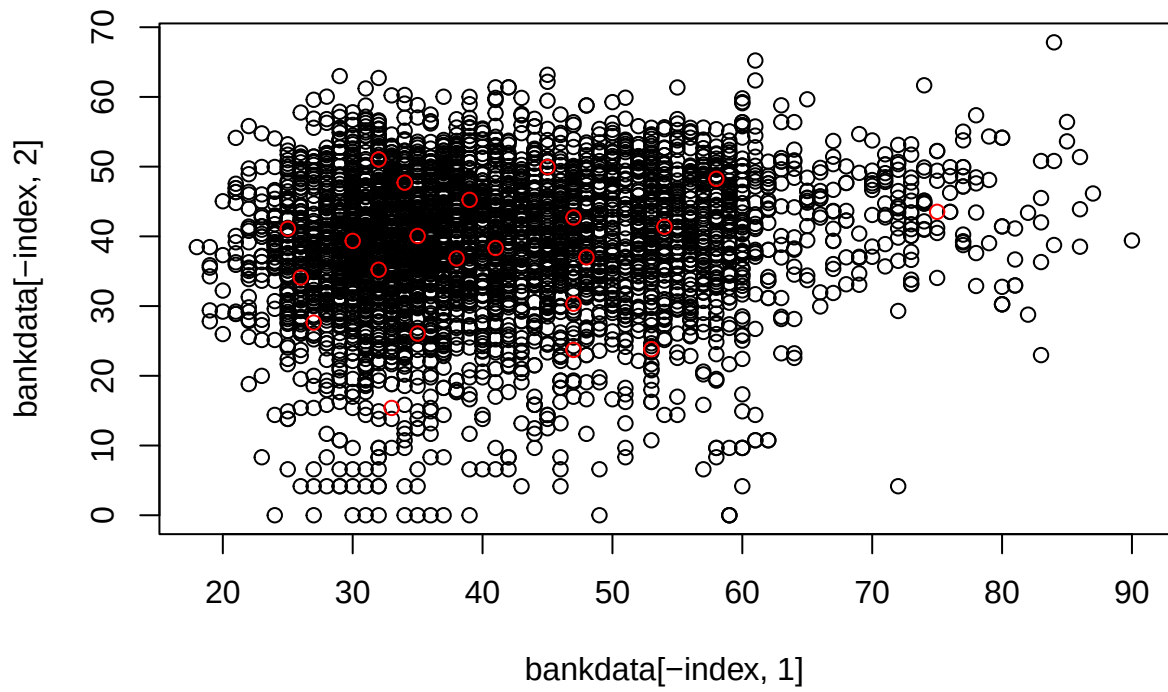


```
## [1] 902 24 1185 550 1571 967 616 3457 362 3905 1428 4188 93 1101 4345
## [16] 273 1420 1997 2292 1432 1217 205
```

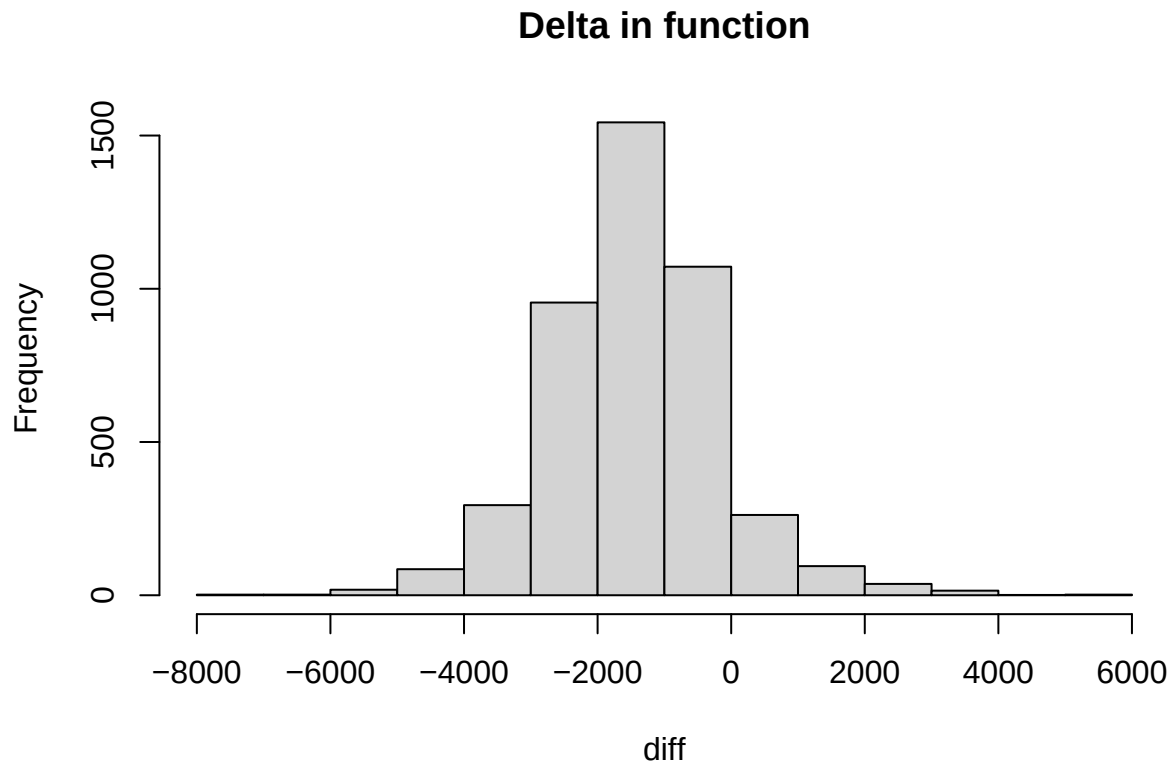
Higher chance of swapping seems to lead to a worse outcome

```
## [1] 10
## [1] 20
## [1] 30
## [1] 40
## [1] 50
## [1] "Starting function value:"
## [1] 21936.02
```

Best individuals tested



```
## [1] "Best function value:"  
## [1] 16434.95  
## [1] "Average diff:"  
## [1] -1443.354
```

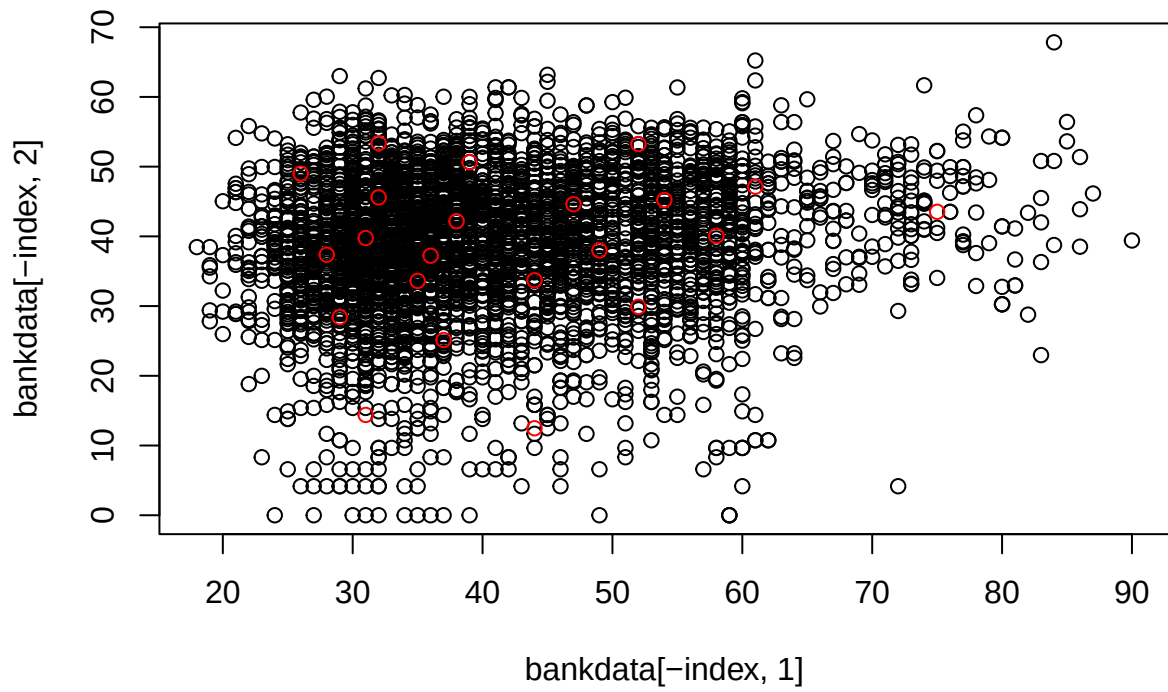



```
## [1] 423 3282 20 4147 411 1470 1301 615 2842 2131 3884 3131 960 444 1747
## [16] 3256 2331 3467 640 4299 1630 1112
```

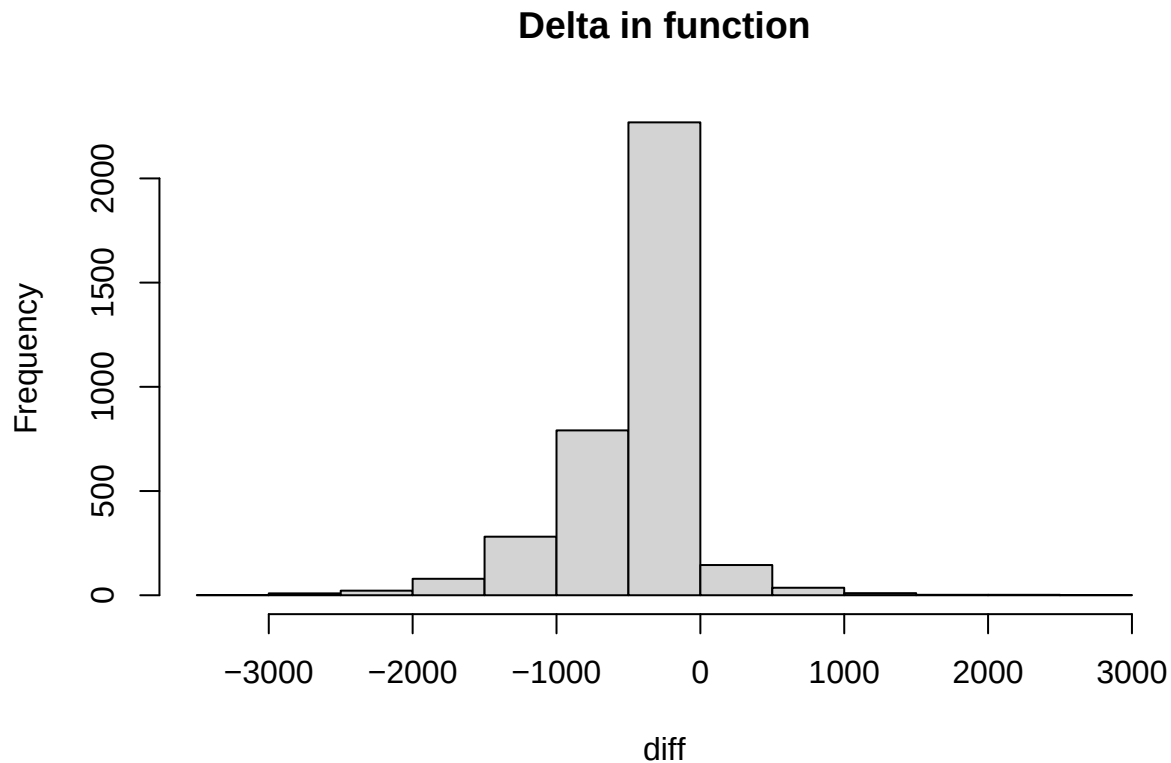
Lower swap rate seems better.

```
## [1] 10
## [1] 20
## [1] 30
## [1] 40
## [1] 50
## [1] "Starting function value:"
## [1] 21936.02
```

Best individuals tested



```
## [1] "Best function value:"  
## [1] 15648.76  
## [1] "Average diff:"  
## [1] -373.4714
```



```
## [1] 3356 4071 1747 2298 968 307 1259 505 1297 1296 2575 3773 369 2057 1421
## [16] 4244 2140 652 1125 1605 1046 255
```

Appendix

```
#####
### Computational statistics, Linköping University, VT2025 ###
### EM algorithm for univariate normal mixtures          ###
### (mixture of two normal distributions)                ###
### Frank Miller                                         ###
#####

emalg <- function(dat, eps=0.000001){

  n      <- length(dat)
  pi_mat <- matrix(NA, nrow=n, ncol=3)

  # define reasonable starting values for parameters; here based on summary statistics for total dataset
  p1 <- 1/3
  p2 <- 1/3
  p3 <- 1/3          # starting value for mixing parameter

  sigma1 <- sd(dat)*2/3      # starting value for standard deviation in group taken as 2/3 of total s
  sigma2 <- sigma1
  sigma3 <- sigma1
```

```

mu1    <- mean(dat) + sigma1/2 # starting values for means, taken a bit lower and a bit higher than o
mu2    <- mean(dat) - sigma1/2
mu3    <- mean(dat)
pv     <- c(p1, p2, p3, mu1, mu2, mu3, sigma1, sigma2, sigma3) # parameter vector

cc     <- eps + 100          # initialize convergence criterion to avoid stopping the while-loop dire

it <- 2

pi_hist <- matrix(1/3, 1, 3)
mu_hist <- matrix(c(mu1, mu2, mu3), 1, 3)
sigma_hist <- matrix(sigma1, 1, 3)

while (cc>eps){
  pv1 <- pv                # save previous parameter vector
  ### E step ###
  for (j in 1:n){
    w1 <- p1 * dnorm(dat[j], mean=mu1, sd=sigma1)
    w2 <- p2 * dnorm(dat[j], mean=mu2, sd=sigma2)
    w3 <- p3 * dnorm(dat[j], mean=mu3, sd=sigma3)

    sum_w <- w1 + w2 + w3
    pi_mat[j, ] <- c(w1, w2, w3) / sum_w

  }
  ### M step ###
  p1 <- mean(pi_mat[, 1])
  p2 <- mean(pi_mat[, 2])
  p3 <- mean(pi_mat[, 3])

  pi_hist <- rbind(pi_hist, c(p1, p2, p3))

  # Update means
  mu1 <- sum(pi_mat[, 1]* dat)/(p1*n)
  mu2 <- sum(pi_mat[, 2]*dat)/(p2*n)
  mu3 <- sum(pi_mat[, 3]*dat)/(p3*n)
  mu_hist <- rbind(mu_hist, c(mu1, mu2, mu3))

  # Update standard deviations
  sigma1 <- sqrt(sum(pi_mat[, 1]*(dat - mu1)^2)/(p1*n))
  sigma2 <- sqrt(sum(pi_mat[, 2]*(dat - mu2)^2)/(p2*n))
  sigma3 <- sqrt(sum(pi_mat[, 3]*(dat - mu3)^2)/(p3*n))
  sigma_hist <- rbind(sigma_hist, c(sigma1, sigma2, sigma3))

  #####
  pv <- c(p1, p2, p3, mu1, mu2, mu3, sigma1, sigma2, sigma3)

```

```

    cossim <- function(v1, v2) {
      t(v2)%*%v1 / sqrt(t(v2)%*%v2 * t(v1)%*%v1)

    }

    cc      <- mean(1-cossim(pv[1:3], pv1[1:3]), 1-cossim(pv[4:6], pv1[4:6]), 1-cossim(pv[7:9], pv1[7:9])

    it <- it + 1
  }

  return(list(pi_hist = pi_hist, mu_hist = mu_hist, sigma_hist = sigma_hist, pv = pv))
}
# example data and run of the algorithm
data <- c(0.1, 0.5, 0.7, 1.1, 2.5, 3.4, 3.5, 3.9, 4.0)

emalg(data)$pv
emalg(data*1000)$pv

data <- load(file = "threepops.Rdata")

emalg(dat3p)$pv

em <- emalg(dat3p, eps=0.000001)

plot_fun <- function(matrix, main) {
  n <- nrow(matrix)
  ylim <- c(min(matrix) * 0.9, max(matrix) * 1.1)

  plot(x = 1:n, y = matrix[, 1], lwd = 2, t = "l", ylim = ylim, main = main)
  lines(x = 1:n,
        y = matrix[, 2],
        col = "blue",
        lwd = 2)
  lines(x = 1:n,
        y = matrix[, 3],
        col = "red",
        lwd = 2)

  legend("topleft", legend = c("Comp 1", "Comp 2", "Comp 3"), col = c("black", "blue", "red"), lwd = 2)
}

plot_fun(em$pi_hist, "pi")
plot_fun(em$mu_hist, "mu")
plot_fun(em$sigma_hist, "sigma")

#####
### Computational statistics, Linköping University, VT2025 ###
### Criterion function Lab 6, Q2 (space filling design)      ###
### Frank Miller                                           ###
###                                                         ###
### We are using in this lab partial data from the         ###

```

```

### original bankdata available at ###
### https://archive.ics.uci.edu/ml/datasets/Bank+Marketing ###
### See also the publication: ###
### Sérgio Moro, P. Cortez, P. Rita (2014). A data-driven ###
### approach to predict the success of bank telemarketing. ###
### Decision Support Systems. ###
#####

# you need to save the following dataset at the right place and/or add/set the path where it is located
load("bankdata.Rdata")

nclients <- dim(bankdata)[1] # number of individuals in the dataset, here 4364

# criterion function: sum of minimal distance to an element in the subset
# dat is the full dataset (here: bankdata), subs is the set of ids for the subset selected
# subs should be a vector of elements in 1, ..., 4364; for this question, it should be of length 22
# example call: crit(bankdata, 1:22), selecting the first 22 individuals
# result of this function is the criterion to be minimized
crit <- function(dat, subs){
  s <- length(subs)
  dist <- matrix(rep(NA, nclients*s), ncol=s)
  for (i in 1:s){
    dist[, i] <- sqrt((dat[,1]-dat[subs[i],1])^2+(dat[,2]-dat[subs[i],2])^2)
  }
  sum(apply(dist, 1, min))
}

# it is good to identify the individuals in the full set by their id (1, ..., 4364),
# then we can sample from this set for the starting subset:
fullset <- 1:nclients

set.seed(123)
index <- sample(fullset, 22)

plot(bankdata[-index, 1], bankdata[-index, 2])
points(bankdata[index, 1], bankdata[index, 2], col = "red")

SA <- function(prob_keep = 0.5, alpha = 0.8, beta = 0.5, temp = 4000, mj = 10, it = 100) {
  set.seed(123)
  index <- sample(fullset, 22)

  diff <- c()

  x_t <- index

  min_function <- crit(bankdata, index)
  min_x <- x_t

  function_x_t <- min_function

```

```

for (j in 1:it) {
  if (j %% (round(it/5)) == 0) {
    print(j)
  }
  for (i in 1:mj) {

    swapped_individuals <- prob_keep < runif(22)

    if (sum(swapped_individuals) == 0) {
      x_star <- x_t
    } else {
      not_swapped <- x_t[!swapped_individuals]
      # swapping individuals that got randomly kicked.
      x_star <- c(not_swapped, sample(setdiff(fullset, x_t), sum(swapped_individuals)))
    }

    function_x_star <- crit(bankdata, x_star)
    if (function_x_star < min_function) {
      min_function <- function_x_star
      min_x <- x_star
    }

    delta <- function_x_t - function_x_star
    hx <- exp(delta/temp)
    diff <- c(diff, delta)

    U <- runif(1)
    if (U < hx) {
      x_t <- x_star
      function_x_t <- function_x_star
      #otherwise we do not accept x_t
    }
  }
  temp <- alpha * temp
  mj <- max(1, floor(beta * mj))
}

print("Starting function value:")
print(crit(bankdata, index))

index <- min_x

plot(bankdata[-index, 1], bankdata[-index, 2], main = "Best individuals tested")
points(bankdata[index, 1], bankdata[index, 2], col = "red")

print("Best function value:")
print(min_function)

print("Average diff:")
print(mean(diff))
hist(diff, main = "Delta in function")

return(min_x)

```

```
}
```

```
# SA(prob_keep = 0.5, alpha = 0.9, beta = 1.02, mj = 50, temp = 10000, it = 200)  
# This gives 16203 as minimum function value but is painfully slow
```

```
set.seed(1234)
```

```
SA(prob_keep = 0.9, alpha = 0.7, beta = 1.03, mj = 50, temp = 100000, it = 50)
```

```
set.seed(1234)
```

```
SA(prob_keep = 0.7, alpha = 0.7, beta = 1.03, mj = 50, temp = 100000, it = 50)
```

```
set.seed(12345)
```

```
SA(prob_keep = 0.95, alpha = 0.7, beta = 1.05, mj = 25, temp = 100000, it = 50)
```